Dr. Wei Guo

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Academic Positions

Hong Kong University of Science and Technology — Sept. 2023 – Present
Postdoctoral Fellow, Department of Electronic and Computer Engineering — Hong Kong SAR, China
Hosts: Prof. Khaled B. Letaief and Prof. Jun Zhang

Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) — Jul. 2023
Ph.D., Computer and Information Engineering. Advisor: Prof. Chuan Huang — Shenzhen, China

University of Electronic Science and Technology of China (UESTC) — Jun. 2017
B.Eng., Communications Engineering — Chengdu, China

Academic Visits

Purple Mountain Laboratories — Sept. 2021 – Nov.2021
Visiting student. — Nanjing, China

The Future Network of Intelligence Institute (FNii), CUHK-Shenzhen — Jun. 2019 – Dec.2019
Visiting student. Advisors: Prof. Chuan Huang and Prof. Shuguang Cui — Shenzhen, China

Shenzhen Research Institute of Big Data (SRIBD) — Feb. 2017 – Jun. 2017
Visiting student. Advisor: Prof. Shuguang Cui — Shenzhen, China

Current Research Interests

Wireless Communications, Generative AI, Integrated Communications and AI

Publications (✉ denotes correspondence)

Journals:

- **W. Guo**, M. He, C. Huang✉, H. He, S. Song, J. Zhang, and K. B. Letaief, "Model-Driven Deep Learning for Distributed Detection with Low-Bit Quantization," *IEEE J. Sel. Areas Commun.* (JSAC), early access, 2026.
- Y. Wang, **W. Guo**✉, C. Huang✉, G. Chen, Y. Zhang, and S. Cui, "Point Cloud Environment-Based Channel Knowledge Map Construction," *IEEE Trans. Cogn. Commun. Netw.*, vol. 12, pp. 9263-9278, 2026.
- J. Tong, **W. Guo**, J. Shao, Q. Wu, Z. Li, Z. Lin, and J. Zhang✉, "WirelessAgent: Large Language Model Agents for Intelligent Wireless Networks," *China Commun.*, vol. 23, no. 3, pp. 265-285, Mar. 2026.
- T. Liao, **W. Guo**✉, H. He, S. Song, J. Zhang, and K. B. Letaief, "Joint Beamforming and Antenna Position Optimization for Fluid Antenna Assisted MU-MIMO Networks," *IEEE J. Sel. Areas Commun.* (JSAC), vol. 44, pp. 1209-1226, 2026.

- J. Shao, J. Tong, Q. Wu, **W. Guo**, Z. Li, Z. Lin, and J. Zhang✉, "WirelessLLM: Empowering Large Language Models Towards Wireless Intelligence," in *J. Commun. Inf. Netw.* (JCIN), vol. 9, no. 2, pp. 99-112, Jun. 2024. **Best Paper Award (2026)**
- **W. Guo**, C. Huang✉, X. Qin, L. Yang, and W. Zhang, "Dynamic Clustering and Power Control for Two-Tier Wireless Federated Learning," in *IEEE Trans. Wireless Commun.* (TWC), vol. 23, no. 2, pp. 1356-1371, Feb. 2024.
- **W. Guo**, R. Li, C. Huang✉, X. Qin, K. Shen, and W. Zhang, "Joint Device Selection and Power Control for Wireless Federated Learning," in *IEEE J. Sel. Areas Commun.* (JSAC), vol. 40, no. 8, pp. 2395-2410, Aug. 2022.
- **W. Guo**, H. Zhang✉, and C. Huang, "Energy Efficiency of Two-Way Communications under Various Duplex Modes," in *IEEE Internet Things J.* (IoT-J), vol. 8, no. 3, pp. 1921-1933, Feb. 2021.

Magazines:

- B. Huang, Y. Wang, **W. Guo**✉, Z. Guo, D. Wu, R. Li, D. Liu, W. Lan, C. Huang✉, and S. Cui, "Building Low-Altitude Communication Networks: A Digital Twin-Based Optimization Framework," *IEEE Wireless Commun. Mag.*, early access, 2026.
- Y. Wang, C. Huang✉, S. Zhang, G. Chen, **W. Guo**✉, S. Lan, L. Xu, X. Cheng, X. Tang, and S. Cui, "Towards Precise Channel Knowledge Map: Exploiting Environmental Information from 2D Visuals to 3D Point Clouds," *IEEE Commun. Mag.*, early access, 2026.

Conference Proceedings:

- Y. Bian, **W. Guo**, Z. Wang, S. Song, J. Zhang, and K. B. Letaief, "GSBF: Gaussian Splatting for Environment-Aware Beamforming," in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, Dec. 2026. **Best Paper Award**
- S. Xie, **W. Guo**, S. Song, J. Zhang, Y.-J. Zhang, and K. B. Letaief, "Hybrid Precision Task-Oriented Communication for OFDM-Based Edge Inference Systems," in *Proc. IEEE Mediterranean Conf. Commun. Netw. (MeditCom)*, Jul. 2026.
- T. Liao, **W. Guo**, J. Qian, S. Song, J. Zhang, and K. B. Letaief, "Rethinking Mutual Coupling in Movable Antenna MIMO Systems," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2026.
- R. Li, Z. Xu, **W. Guo**, and Y.-J. Zhang, "Divergence Meets Dispersion: Efficient Wideband Beam Training for XL-MIMO," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2026. **Best Paper Award**
- Y. Bian, **W. Guo**, J. Yang, S. Song, J. Zhang, S. Jin, and K. B. Letaief, "Multi-modal Data Driven Virtual Base Station Construction for Massive MIMO Beam Alignment," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2026.
- E. Tang, **W. Guo**, H. He, S. Song, J. Zhang, and K. B. Letaief, "Accurate and Fast Channel Estimation for Fluid Antenna Systems with Diffusion Models," in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, pp. 2066-2071, Dec. 2025.
- T. Liao, **W. Guo**, H. He, S. Song, J. Zhang, and K. B. Letaief, "Fluid Antenna-Assisted MU-MIMO Systems with Decentralized Baseband Processing," in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, pp. 2372-2377, Dec. 2025.
- **W. Guo**, M. He, C. Huang, H. He, S. Song, J. Zhang, and K. B. Letaief, "Model-driven Deep Learning for Distributed Detection in WSNs with Binary Quantization," in *Proc. IEEE Mediterranean Conf. Commun. Netw. (MeditCom)*, Jul. 2024.

- **W. Guo**, R. Li, C. Huang, X. Qin, K. Shen, and W. Zhang, "Optimized Device Selection and Power Control for Wireless Federated Learning," in *Proc. IEEE Glob. Commun. Conf. (Globecom)*, Dec. 2022.
- **W. Guo**, C. Huang, X. Qin, L. Yang, and W. Zhang, "Dynamic Clustering and Power Control for Two-Tier Over-the-Air Federated Learning," in *Proc. IEEE/CIC Int. Conf. Commun. China (ICCC)*, Aug. 2022.
- **W. Guo** and C. Huang, "Energy Efficiency of Full-Duplex Two-Way Channels," in *Proc. Glob. Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2019.

Patents:

- CN patent: "一种面向无线联邦学习的用户协同传输方法", 黄川, 崔曙光, **郭玮**. 专利申请号: 202211032064.0, 2022.

Preprints:

- **W. Guo**, T. Liao, Z. Wang, S. Song, J. Zhang, and K. B. Letaief, "Mutual-Coupling-Aware Joint Precoding and Antenna Positioning for Movable-Antenna MU-MIMO Systems," arXiv: 2611.04552, 2026.
- **W. Guo**, Y. Zhang, X. Ge, and J. Zhang, "Equal Path Cost, Unequal Output Effects: Understanding Perturbation Propagation in Diffusion Models," arXiv: 2610.11380, 2026.
- Y. Bian, K. Zhang, **W. Guo**, Z. Wang, S. Song, J. Zhang, and K. B. Letaief, "AIMS: An Agentic AI Framework for Sim-to-Real Multi-Modal ISAC," arXiv: 2609.39964, 2026.
- S. Xie, **W. Guo**✉, S. Song, J. Zhang, Y.-J. Zhang, K. B. Letaief, "Task-Oriented Communication with Hybrid-Precision Models", arXiv: 2607.16766, 2026, submitted to *IEEE Trans. Mobile Comput.*, currently under **major revision**.
- Y. Bian, **W. Guo**✉, J. Yang, S. Song, J. Zhang, S. Jin, and K. B. Letaief, "Multi-Modal Environment-Aware Beam Management for Massive MIMO: A Geometry-Driven Virtual Base Station Framework," arXiv: 2606.26567, 2026, submitted to *IEEE Trans. Wireless Commun.*, currently under **major revision**.
- E. Tang, **W. Guo**✉, H. He, S. Song, J. Zhang, K. B. Letaief, "A Unified Two-Stage Generative Diffusion Framework for Channel Estimation and Port Selection in Multiuser MIMO-FAS", arXiv: 2605.29679, 2026, submitted to *IEEE Trans. Wireless Commun.*, currently under **major revision**.
- T. Liao, **W. Guo**✉, J. Qian, Z. Wang, S. Song, J. Zhang, and K. B. Letaief, "Rethinking Mutual Coupling in Movable Antenna MIMO Systems: Modeling and Optimization," arXiv: 2604.26282, 2026, submitted to *IEEE Trans. Wireless Commun.*
- J. Tong, Z. Li, F. Liu, **W. Guo**, and J. Zhang✉, "WirelessAgent++: Automated Agentic Workflow Design and Benchmarking for Wireless Networks," arXiv: 2603.00501, 2026, submitted to *IEEE Trans. Wireless Commun.*, currently under **major revision**.

Awards

IEEE GLOBECOM Best Paper Award	2026
Journal of Communications and Information Networks (JCIN) Best Paper Award	2026
IEEE WCNC Best Paper Award	2026
Exemplary Reviewer of IEEE Wireless Communications Letters	2020

Teachings

Teaching assistant at CUHK-Shenzhen:

EIE4006 Performances Evaluation of Communication Networks	Sum' 2020-21
CIE6126 Performances Analysis of Computer and Communication Networks	Sum' 2020-21
EIE3001 Signals and Systems	Spr' 2021

Academic Services

Conference Reviewers:

- IEEE Global Communications Conference (Globecom)
- IEEE/CIC International Conference on Communications in China (ICCC)
- IEEE Wireless Communications and Networking Conference (WCNC)
- IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)
- ...

Journal Reviewers:

- IEEE Transactions on Wireless Communications
- IEEE Transactions on Communications
- IEEE Journal on Selected Areas in Information Theory
- IEEE Wireless Communications Letters
- IEEE Systems Journal
- IEEE Transactions on Machine Learning in Communications and Networking
- IEEE Internet of Things Magazine
- China Communications
- Journal of Communications and Information Networks
- ...

Presentations

2024 IEEE Hong Kong 6G Wireless Summit (HK6GWS), Hong Kong, Poster presentation
2024 IEEE MeditCom, Madrid, Spain, Oral presentation
2022 IEEE Globecom, Rio de Janeiro, Brazil, virtual presentation
2022 IEEE/CIC ICCC, Foshan, China, Oral presentation
2019 IEEE GlobalSIP, Ottawa, Canada, Oral presentation